

Week-3 Essay

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Introduction

In an age where data increasingly guides decision-making, Cathy O'Neil's *Weapons of Math Destruction* chapter five and Seth Stephens-Davidowitz's *Everybody Lies* chapter three offer critical perspectives on how algorithms and big data shape our lives. O'Neil explains how algorithms in criminal justice can worsen bias and inequality, focusing more on efficiency than fairness. Meanwhile, Stephens-Davidowitz shows how big data can uncover deeper, more honest truths about human behavior that traditional methods often miss. This essay will explore how both authors highlight the power of data, while also warning about its potential dangers when used wrongly.

Literature Review

In Chapter 5 of *Weapons of Math Destruction* the author Cathy O'Neil reviews the use of algorithms in criminal justice, where they are already being employed for "predictive policing" and "recidivism risk" assessments (O'Neil, 2016). At their best, these tools claim to add efficiency and objectivity in deciding who is likely to commit a crime. At their worst - to use O'Neil's clever phrase these are the views of jackals snarling out of the bushes. O'Neil also symmetrically demonstrates her commitment to civil rights by taking examples of algorithms that end up being used against both blacks and Hispanics to rent serve as home loans in poor neighborhoods and to recruit the select kind of people who might fit into "golden neighborhoods" (O'Neil, 2016). She points out that these models create a feedback loop. As law enforcement adopts predictive policing tools, it increases surveillance in certain neighborhoods and subsequently engages in more arrests -- thereby reinforcing the cycle of over policing on those streets. Author argues that algorithms used in the justice system often reinforce bias, labeling entire communities as "high risk" based on flawed data. These "*Weapons of Math Destruction*" prioritize efficiency over fairness, making it hard to challenge decisions like bail, sentence (O'Neil, 2016). Instead of promoting safety, these tools can worsen inequality, punishing people more for where they come from or their background than for their actual behavior.

In Chapter 3, "*Data Reimagined*", Seth Stephens-Davidowitz explores the transformative power of big data, specifically how it allows us to reimagine the ways we understand human behavior (Davidowitz, 2017). He argues that traditional methods of data collection, like surveys or polls, are often inaccurate because they rely on people's willingness to be truthful or self-aware. However, data from online platforms particularly Google searches provides a more honest reflection of people's thoughts and desires (Davidowitz, 2017). Stephens-Davidowitz highlights how "big data enables us to peer into the collective mind," revealing hidden truths and contradictions in human nature. He discusses several examples where big data offers surprising insights, such as how search trends can reveal societal prejudices, health concerns, or personal struggles that people may not openly discuss (Davidowitz, 2017). By comparing online behavior to traditional social science methods, the chapter emphasizes how data from the digital world can serve as a lens for understanding the real world, challenging old assumptions and providing a more nuanced picture of what people actually think and feel.

Analysis of the Issue

Both authors discuss the growing influence of data, but with different concerns. O'Neil warns that algorithms, particularly in criminal justice, may increase racial bias and inequality by prioritizing productivity over fairness. In contrast, Stephens-Davidowitz believes that big data can reveal more fundamental truths about human behavior, offering more honest insights than traditional methods. However, he also cautions that big data can be misused or misinterpreted.

But both authors agree that, when used irresponsibly, data can perpetuate harmful biases. O'Neil focuses on the dangers of biased algorithms, while Stephens-Davidowitz highlights the potential of big data to uncover truths, but also its risks of manipulation. Lastly, both highlight that how data is used is extremely important.

Conclusion and Limitations

O'Neil and Stephens-Davidowitz offer important insights into the power and risks of data. O'Neil warns about the dangers of biased algorithms in criminal justice, while Stephens-Davidowitz celebrates the potential of big data to reveal hidden truths. Together, they emphasize the need for careful, ethical use of data to avoid reinforcing inequalities and ensure that technology serves society fairly and responsibly. From my perspective, the key takeaway is the need for responsible and ethical use of data. While data itself is neutral, how we collect, interpret, and apply it can either create positive change or reinforce injustice. Both O'Neil and Stephens-Davidowitz argue for more transparency and accountability in data-driven systems. We must be aware of how data is used, ensuring that its power is harnessed for fairness and equality rather than reinforcing systemic biases.

References

- Davidowitz, S. S. (2017). *Everybody lies: Big data, new data, and what the internet can tell us about who we really are*. Dey Street Books.
- O'Neil, C. (2016). In *Weapons of math destruction: how big data increases inequality and threatens democracy*. Crown Publishers.